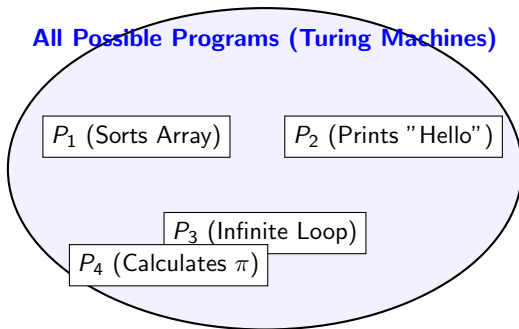


Rice's Theorem

A Visual Perspective

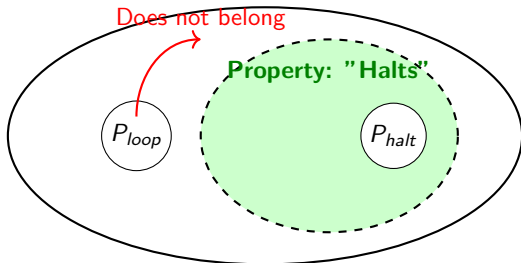
Theory of Computation

1. The Universe of Programs



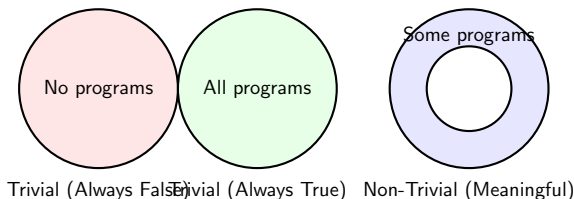
- Imagine a giant circle containing every program ever written or that could be written.

2. What is a "Property"?



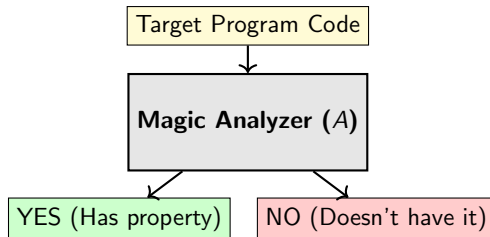
- A **Property** is just a behavior. It divides our universe into two sets: Programs that do it, and programs that don't.

3. Trivial vs. Non-Trivial Properties



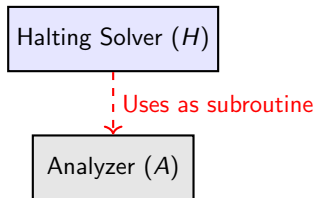
- **Trivial:** Applies to ALL programs or NO programs.
- **Non-Trivial:** Applies to SOME programs but not others.

4. The Dream: The Magic Analyzer



- We want a program (A) that reads another program's code and tells us if it has a non-trivial property (e.g., "Is it malware?").

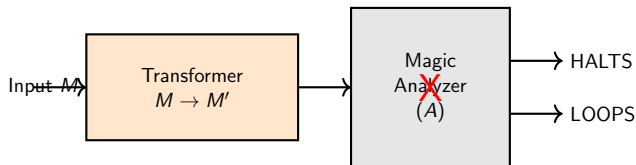
5. The Halting Problem Trap



If A exists, H exists!

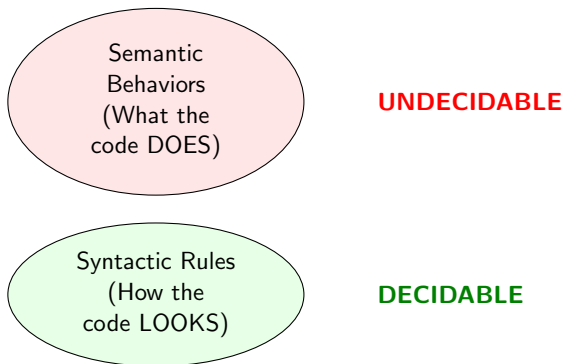
- Alan Turing proved we cannot build H (The Halting Solver).
- If our Magic Analyzer (A) existed, we could use it to secretly build H .

6. The Impossible Reduction



- By transforming any program into a special test program (M'), the Analyzer A is forced to solve the Halting Problem.

7. What this means visually



- We can write algorithms to check *syntax* (e.g., "Does it have a while loop?").
- We **cannot** write algorithms to check *semantics* (e.g., "Will it print 5?").

8. Conclusion of Rice's Theorem

ALL non-trivial semantic properties of programs are **undecidable**.

- There is no perfect anti-virus.
- There is no perfect bug-finder.
- Static analysis tools will always be approximations!